



Basics of Atomic Theories & Particle Physics

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Course: Applied Physics

Introduction to Atomic Theories

1897: J.J. Thomson

Discovers the electron, the first subatomic particle, revolutionizing the understanding of atomic structure.

1

2

1911: Ernest Rutherford

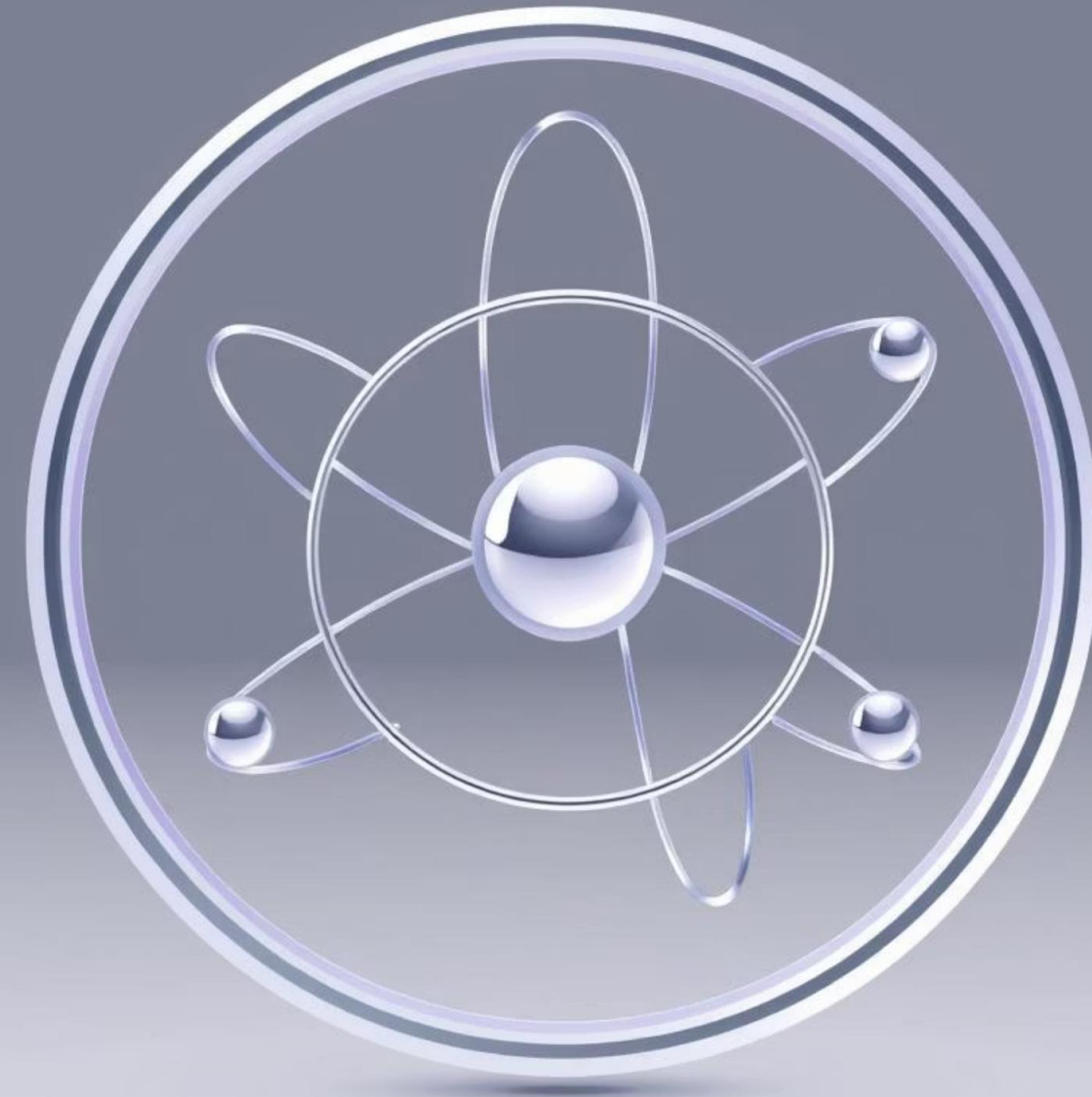
His gold foil experiment reveals the existence of the atomic nucleus, a dense, positively charged core.

3

1932: James Chadwick

Discovers the neutron, a neutral particle within the nucleus, completing the basic atomic model.

Atoms are now understood to consist of electrons orbiting a nucleus made of protons and neutrons.



From Atoms to Particles: The Birth of Particle Physics

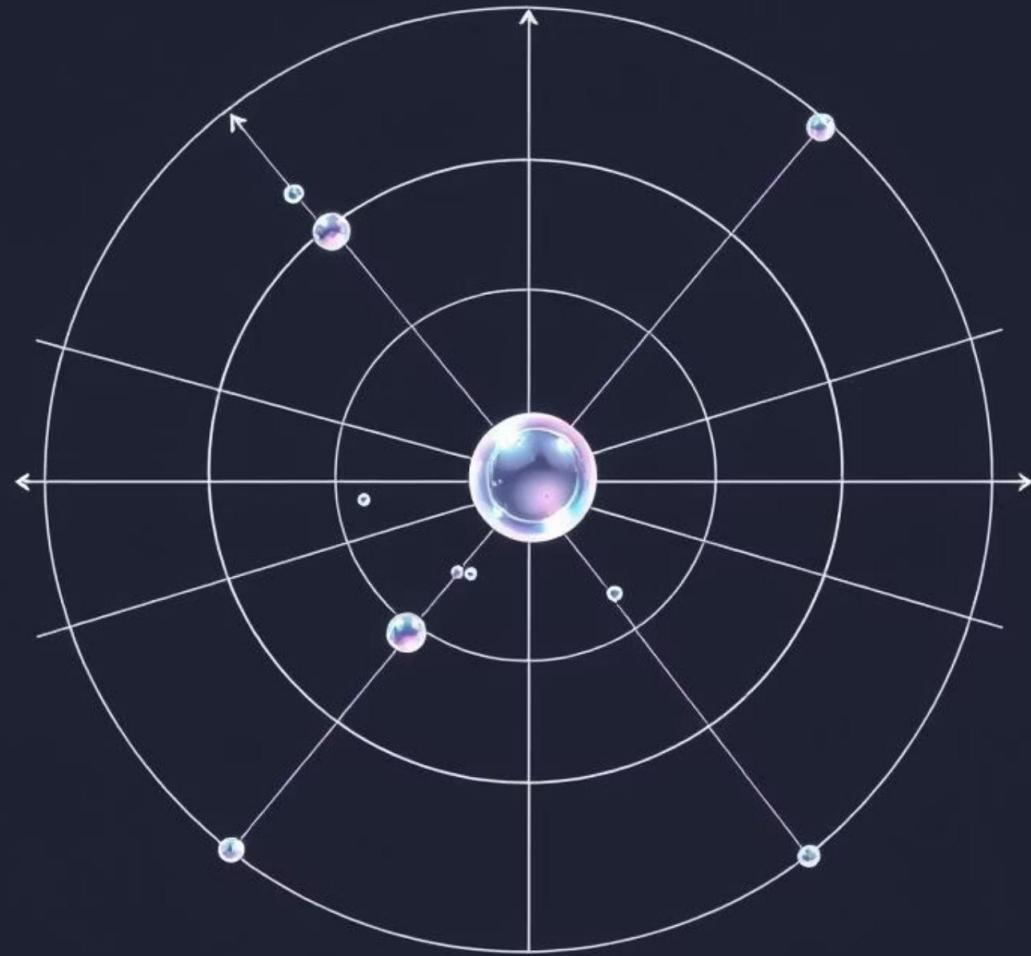
The early 20th century saw the discovery of many new particles beyond the familiar protons, neutrons, and electrons.

This necessitated a unified theory to explain these fundamental particles and the forces governing their interactions.

Particle accelerators simulate conditions just after the Big Bang, allowing scientists to study these elusive particles.



The Standard Model: A Modern Framework



1

Origins

Developed collaboratively by many physicists in the 1960s-70s, it represents our most comprehensive theory of particle physics.

2

Particles

It describes 17 fundamental particles: 12 fermions (matter particles) and 5 bosons (force carriers).

3

Forces

Explains three fundamental forces: electromagnetic, strong nuclear, and weak nuclear forces.

4

Limitations

Crucially, gravity remains outside the Standard Model's scope, posing one of physics' greatest challenges.

The 17 Fundamental Particles

- **Fermions (Matter Particles):**
 - **6 Quarks:** Up, Down, Charm, Strange, Top, Bottom (constituents of protons and neutrons)
 - **6 Leptons:** Electron, Muon, Tau, and their corresponding Neutrinos
- **Bosons (Force Carriers):**
 - **Photon:** Mediates electromagnetism
 - **Gluon:** Mediates the strong nuclear force
 - **W & Z Bosons:** Mediate the weak nuclear force
 - **Higgs Boson:** Responsible for giving other particles mass

Protons and neutrons are not fundamental; they are composite particles made of quarks.

The monumental discovery of the Higgs boson in 2012 at CERN confirmed the mechanism by which particles acquire mass.



Mathematical Formula: The Standard Model Lagrangian

The Standard Model's core is elegantly captured in its Lagrangian, a concise mathematical expression that encodes all known fundamental particle interactions and dynamics.

$$\mathcal{L}_{\text{SM}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + i\bar{\psi}\not{D}\psi + |D_{\mu}H|^2 - V(|H|^2) + \text{Yukawa terms}$$



Strong Force Interactions

Encoded through terms describing gluons and the "color charge" of quarks, dictating how they bind together.



Electroweak Unification

Describes the combined electromagnetic and weak forces via W, Z bosons, and photons.



Mass Generation

Includes terms for matter particles' masses, arising from their interaction with the omnipresent Higgs field.

Despite its apparent complexity, this formula predicts particle behavior with extraordinary precision, matching experimental results to an unprecedented degree.

Visualizing the Standard Model

This comprehensive chart of the Standard Model particles illustrates:

- **Fermions:** Arranged in three generations, showing increasing mass from the first to the third generation.
- **Bosons:** Depicted as the fundamental force carriers, mediating interactions between matter particles.
- **Higgs Boson:** Highlighted as the unique particle responsible for imparting mass to other fundamental particles.

Standard Model

Qp Quarks	I Borark	He Durark	Rs Langers	Sd Plactds	
Sc Quarks	Dp Fossets	Qa Desons	Sp Leptons	Re Bosons	
Qu Scrattls	Bμ Bosons	Re Bosons	He Bosons	Ce Bosons	
Be Mesus	Cr Leptork	Ie Plessus	Qd Bosens	Cs Sierack	Ca Bosons

Applications of the Standard Model



Nuclear Reactions

Explains processes like nuclear fusion powering the Sun, driven by the weak nuclear force.



Technological Foundation

Serves as the theoretical bedrock for technologies such as PET scans and advanced particle detectors.



Accelerator Guidance

Guides research at cutting-edge particle accelerators like the Large Hadron Collider (LHC) at CERN.



Beyond Standard Model Physics

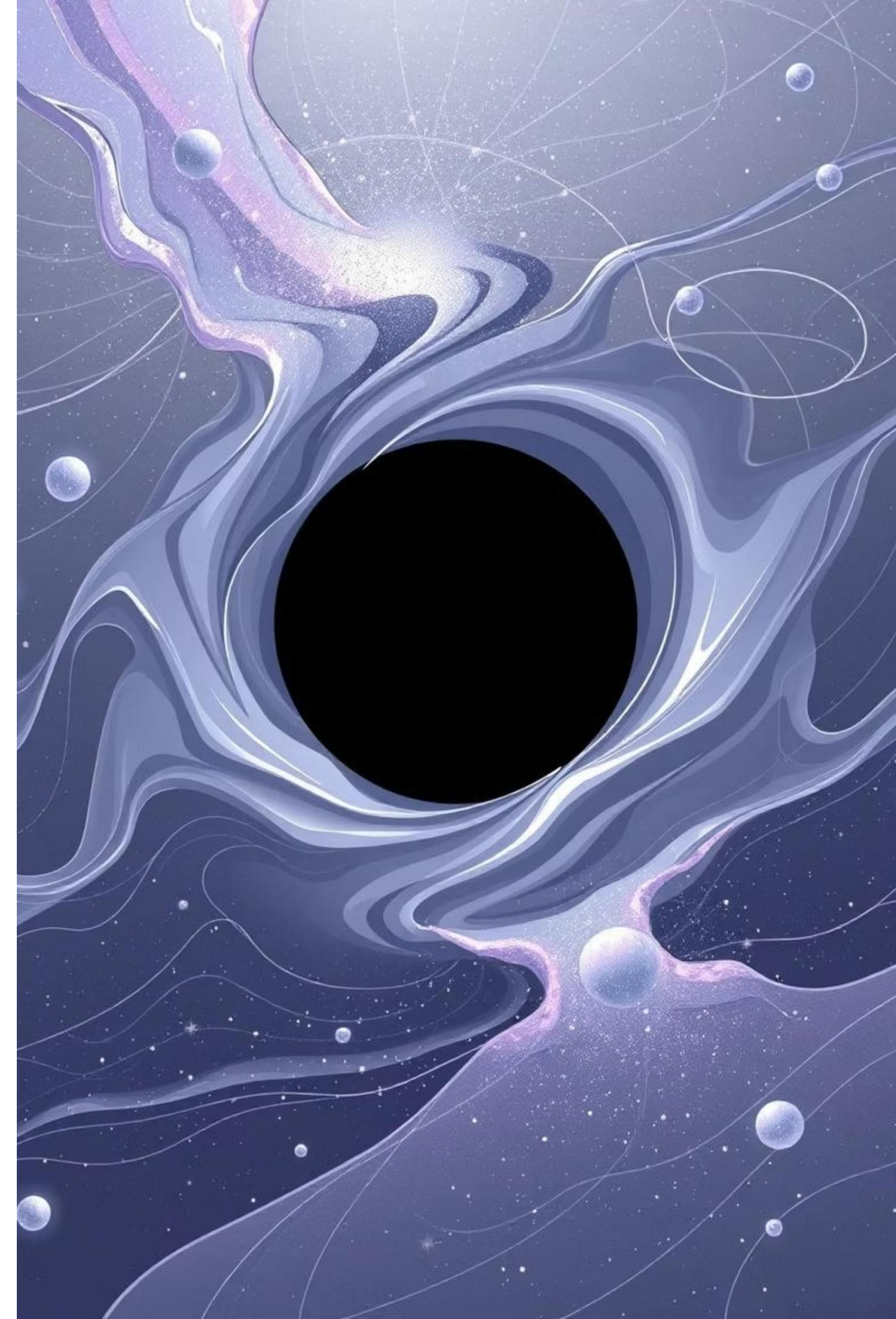
Helps physicists formulate hypotheses and search for new physics, including dark matter and neutrino masses.

Current Frontiers & Open Questions

The Standard Model is incredibly successful, but it doesn't answer everything. Here are some of the universe's most profound unsolved puzzles:

- **The Enigma of Gravity:** Why does gravity, a fundamental force, remain outside the Standard Model's descriptive power?
- **Dark Matter & Dark Energy:** What constitutes the mysterious dark matter and dark energy that account for 95% of the universe's mass-energy budget?
- **Neutrino Mass:** Do neutrinos acquire mass solely via the Higgs field, or is there another, yet undiscovered, mechanism at play?

Ongoing experiments and theoretical developments are relentlessly pursuing these questions, aiming to unify all fundamental forces and discover new particles.



Conclusion: The Standard Model's Legacy and Future

The Standard Model stands as a monumental triumph of 20th-century physics, successfully explaining the fundamental building blocks of the universe and their interactions.



- It continues to inspire groundbreaking discoveries and drive the development of new technologies.
- The relentless quest to complete our understanding of the cosmos fuels modern physics forward.

Thank you!